Final

Cassidy Thorpe

2026-06-09

Set up

```
# load in packages
library(tidyverse)
library(here)
library(janitor)
library(DHARMa)
library(ggeffects)
# load in data
nest <- read_csv(here("data","nest_data_final.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used a logistic regression. In part 2, they used a chi-square test.

b. Suggestions for rewriting

Part 1: These results indicate that the distance to the water's edge has an influence on whether on not the American Avocets will use a microhabitat.

Part 2: These results indicate that different species of birds such as the American Avocet, Forester's Tern, and Black-necked Stilt favor different habitat types.

c. Further analysis

Another variable mentioned in the paper is the American Avocet's access to islands for breeding and roosting and how far away the nesting island is. The distance to the nesting island is a continuous variable, measured in meters, and it was found that the Avocets were on average, 131 meters closer to the nesting island than expected. Access to these island is important for the rest and reproduction of American Avocets. If the birds are closer to the islands, they may choose to occupy that microhabitat over their other options since they seem to like them.

Problem 2. Data analysis

a. Clean and wrangle

```
# create new object from nest
nests_clean <- nest |>
  # select columns of interest (the columns I want to answer the research
  # question are height, ant species, nest occurrence)
  select(height_cm, ant_species, case_control) |>
  # filter for specific species
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  #
  mutate(
    # reorder the ant species
    ant_species = fct_relevel(ant_species, "AB", "RRB", "BBR"),
    # add a column for the scientific names
    scientific_name = recode_values(ant_species,
                                    "AB"  ~ "Crematogaster sjostedti",
                                    "RRB" ~ "Crematogaster mimosae",
                                    "BBR" ~ "Crematogaster nigriceps"))
# display structure of data frame
glimpse(nests_clean)
```

```
Rows: 270
Columns: 4
$ height_cm       <dbl> 391.5, 310.0, 513.0, 154.0, 324.5, 413.0, 167.0, 276.5~
$ ant_species     <fct> RRB, RRB, RRB, RRB, AB, RRB, BBR, RRB, AB, BBR, RRB, R~
$ case_control    <dbl> 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, ~
$ scientific_name <chr> "Crematogaster mimosae", "Crematogaster mimosae", "Cre~
```

```r
# display 10 random rows
nests_clean |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 4
   height_cm ant_species case_control scientific_name
       <dbl> <fct>              <dbl> <chr>
 1      212. RRB                    1 Crematogaster mimosae
 2      276. RRB                    0 Crematogaster mimosae
 3      298. RRB                    0 Crematogaster mimosae
 4      134. RRB                    0 Crematogaster mimosae
 5       90  RRB                    0 Crematogaster mimosae
 6      236. RRB                    1 Crematogaster mimosae
 7      114. RRB                    0 Crematogaster mimosae
 8      247  RRB                    1 Crematogaster mimosae
 9      130. AB                     0 Crematogaster sjostedti
10      238. RRB                    1 Crematogaster mimosae
```

b. Response variable

The 1s mean that the tree contained a nest and the 0s mean that the tree did not contain a nest.

c. Purpose of the study

Tree height: Tree height is a continuous variable because a tree can be any height measurement. As tree height increases, the probability of nest occurrence could increase because taller trees that are older may have thicker branches and more space to nest. This information is based on what I read in the results section in paragraphs 1 and 2 as well as the discussion section in paragraphs 1 and 2.

Acacia ant species: Acacia ant species is a categorical variable because it groups the trees by ant symbiont species. The tree with the more aggressive species of ants had a higher probability of having a bird nest because the ants are more likely to deter predators that might want to climb the tree to get to the bird nest. This is based on information I read in the results section in paragraphs 1 and 2.

d. Table of models

Model number	Tree height	Ant species	Interaction (Height × Species)	Predictor list
1	X	X		tree height and ant species
2	X	X	X	tree height, ant species, and their interaction

e. Run the models

```
# model 1: without interaction
model1 <- glm(
  case_control ~ height_cm + ant_species,
  data = nests_clean,
  family = binomial
)

# model 2: with interaction
model2 <- glm(
  case_control ~ height_cm * ant_species,
  data = nests_clean,
  family = binomial)
```

f. Select the best model

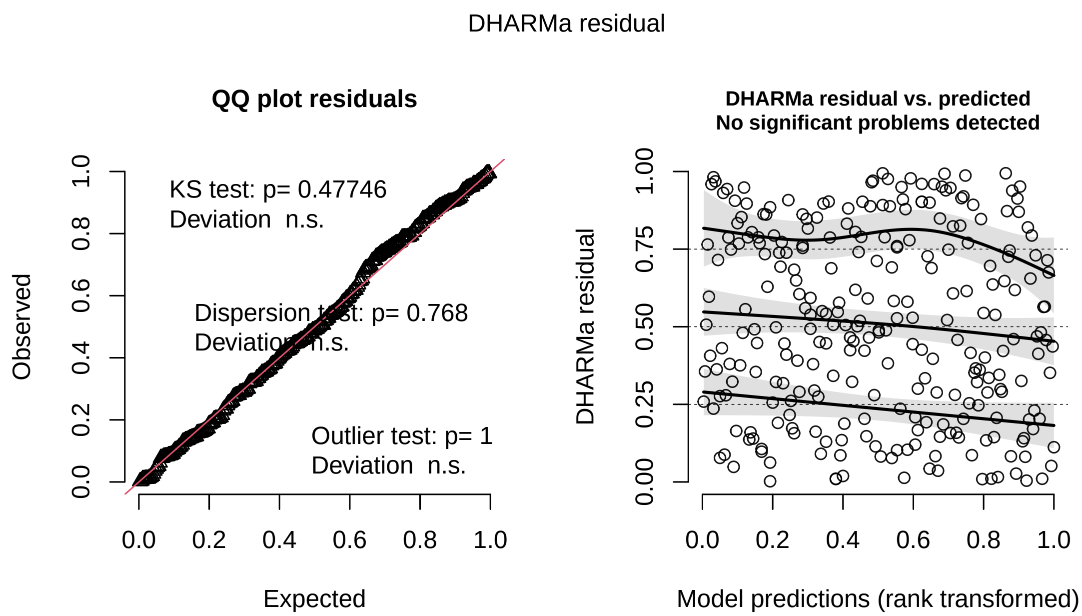
```
# compare models
AIC(model1, model2)
```

```
       df      AIC
model1  4 258.7430
model2  6 237.8042
```

The best model was model 2, the model with the interaction. This model includes the tree height, acacia ant species, and the interaction between them. This tells us if the relationship between tree height and the probability of a nest changes based on the ant species.

g. Check diagnostics

```
# simulate the residuals - use the selected model
simulated_res <- simulateResiduals(fittedModel = model2, n = 250)
# plot the diagnostic outputs
plot(simulated_res)
```

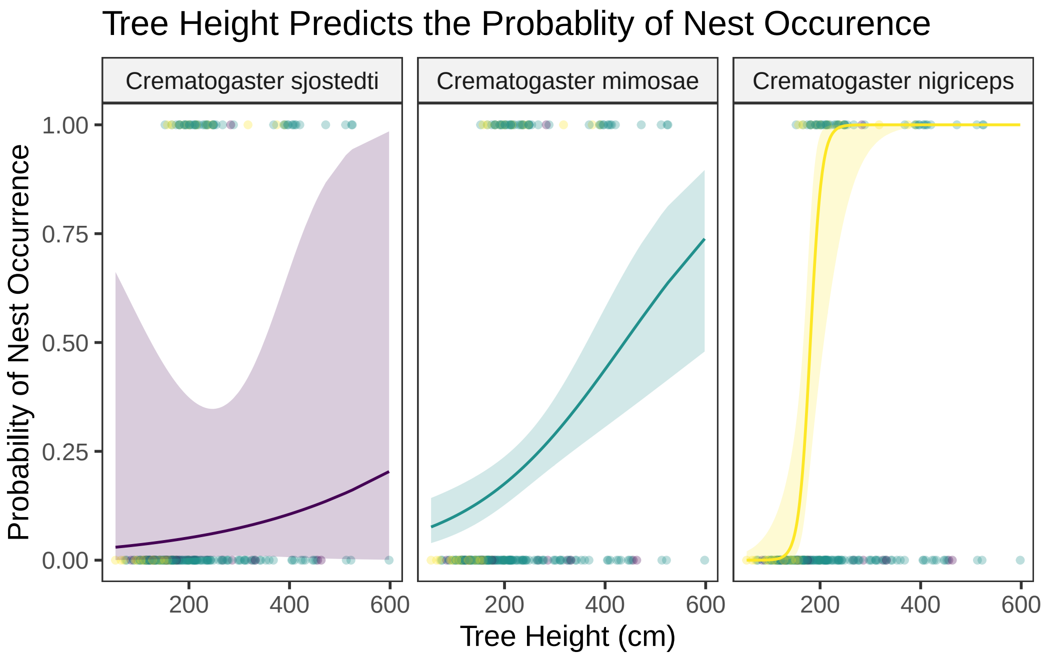


- Residuals are normally distributed, the points follow the red line (QQ plot)
- Residuals are homoscedastic, the points look random (DHARMa residuals vs. predicted)

h. Visualize model predictions

```
# extract predicted probabilities along the range of tree heights for each ant species
# [all] means look at every measured height
predictions <- ggpredict(model2, terms = c("height_cm [all]", "ant_species"))
ant_names <- c(
  "RRB"   = "Crematogaster mimosae",
  "BBR" = "Crematogaster nigriceps",
  "AB"  = "Crematogaster sjostedti")
# base layer: ggplot
```

```r
ggplot() +
  geom_jitter(
    data = nests_clean,
    aes(x = height_cm,
        y = case_control,
        color = ant_species),
    width = 0.1,
    height = 0,
    alpha = 0.3,
    size = 0.9) +
  geom_line(
    data = predictions,
    aes(x = x,
        y = predicted,
        color = group)) +
  geom_ribbon(
    data = predictions,
    aes(x = x,
        ymin = conf.low,
        ymax = conf.high,
        fill = group),
    alpha = 0.2) +
  facet_wrap(~ group,labeller = as_labeller(ant_names)) +
  labs(
    x = "Tree Height (cm)",
    y = "Probability of Nest Occurrence",
    title = "Tree Height Predicts the Probablity of Nest Occurence") +
  scale_color_viridis_d() +
  scale_fill_viridis_d() +
  theme_bw() +
  theme(
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    legend.position = "none",
    strip.background = element_rect(fill = "grey95"))
```


Tree Height Predicts the Probablity of Nest Occurence
Crematogaster sjostedti
Crematogaster mimosae
Crematogaster nigriceps
Probability of Nest Occurrence
1.00
0.75
0.50
0.25
0.00
200
400
600
Tree Height (cm)